

Trainings ▾

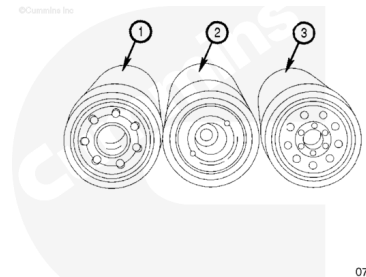
Remove

⚠ CAUTION ⚠

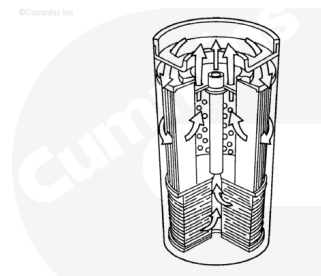
The bypass and combination filters both have the same threads. Verify correct filter is used for replacement to avoid damage to the engine.

The external appearance of the full-flow (1), the bypass (2), and the combination (3) filters are the same. The accompanying picture identifies the differences among the three filters.

Note : The full-flow filter contains [1 1/2-16 in] threads. The bypass and combination filters contain [2 1/4-12 in] threads.



A combination oil filter is used on **most** engines. The upper section of the filter contains the full-flow filter element, while the lower section contains the bypass element.

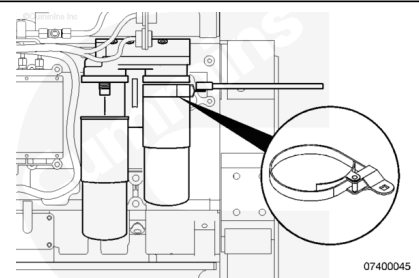


⚠ CAUTION ⚠

Use caution when draining oil or replacing filters that the oil is not spilled or drained into the bilge area. The oil and oil filters must be disposed of in accordance with local environmental regulations.

Note : The following illustrations show the combination oil filter. Use the same procedure when changing the remote bypass oil filters.

Use an oil filter wrench, Part Number 3375049, or equivalent. Remove the oil filters.

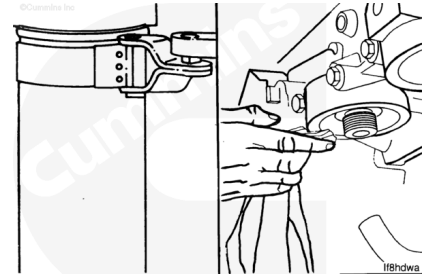


Discard the filters if they are **not** needed for a failure analysis.

Install

Clean the head surface of the new filter.

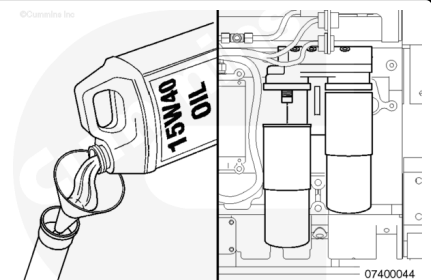
Use engine oil to lubricate the gasket surface of the filters.



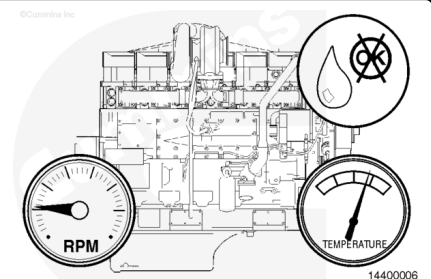
Fill the filters with clean, 15W-40 oil. Install the filters on the oil filter head.

Note : Mechanical overtightening can distort the threads or damage the filter element seal.

Turn the filter until the seal contacts the head. Turn an additional 3/4-turn to 1 turn.

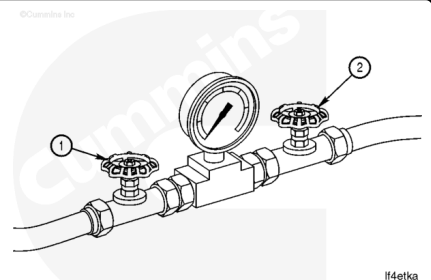


Operate the engine until the oil temperature is at least 65°C [150°F]. Check for leaks.



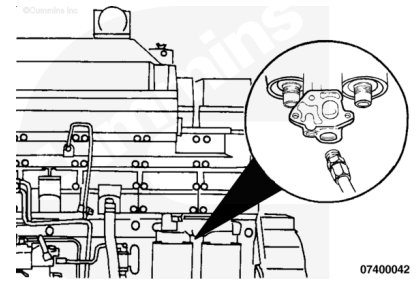
Pressure Differential Test

Use a differential pressure gauge, or one pressure gauge with two oil hoses and two valves to eliminate gauge error. Use a gauge with a minimum pressure capacity of 517 kPa [75 psi].



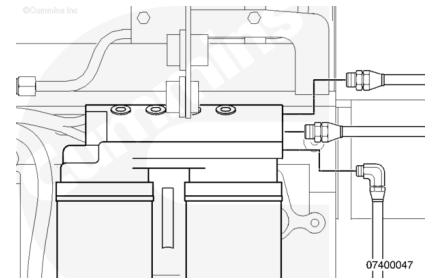
Connect one hose in the [1-5/16-12] SAE plug hole at the bottom of the nose.

This reading will indicate oil pressure before the oil goes through the filters.



Connect one hose in the [1-5/16-12] SAE plug hole on side of engine.

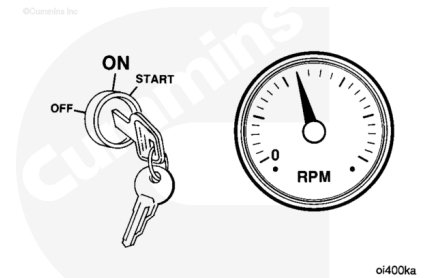
This reading will indicate the oil pressure after the oil goes through the filters.



Operate the engine at rated rpm and normal operating temperature.

Open the valve to read the pressure after the filter first.

Then close the valve and open the other valve to read the oil pressure before the filter.



If the difference in pressure is more than 103 kPa [15 psi], replace the filters.

If the difference in pressure using clean oil filters is more than 69 kPa [10 psi], an oil filter with excessive restriction is being used. Replace the oil filters.

Maximum Oil Filter Pressure Differential: 69 kPa [10 psi]

